

Task 2: Test Log and Iterative Testing Documentation

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Introduction to Testing Approach:

Tests were conducted through the entire development process and not only at its completion. I followed an iterative strategy which included adding one feature after another, quickly running it, fixing all the errors and continuing to the next feature. This allowed me to save a lot of time since I could easily spot and address potential issues early on.

The features I tested included:

Input validation - testing form rejection of erroneous input

Calculation processes - ensuring that loyalty points were calculated properly

Processes in general - testing full user processes from beginning to end

Navigation - testing that all links led to the correct page

Map loading - API testing

Test Plan:

I made a simple test plan before testing of what I needed to check

Tested Functionality	Reason
User Registration	User input validation, password matching, email validation. Check whether invalid information enters the database
Login	Valid username and password login, incorrect login fails. Ensure account security
Product Page	Products display, filters apply, search feature. Check whether customers can find products on the site or not
Shopping Cart and Checkout	Products added to cart, prices calculated, orders placed. Core functions of the e-commerce website
Inventory Management	Inventory reduces after purchase, unable to purchase above stock. Prevents sale of unavailable products
Loyalty Points	Earned after purchase, redeemable for discounts. Loyalty program for customers must function properly.
Producer Dashboard	Only their products listed for producers, updated stock saves. Producers ability to manage products

Map API Integration	Marks display on map, popups work on maps; Show integration with an external AP.
Dark Mode Feature	Change dark mode setting on all pages. Accessibility requirement

Test Log:

Test No.	Function Tested	Input Used	Expected Outcome	Actual Outcome	Result	Action Taken
1	Registration form validation	Empty full name field	Error message appears	Error showed correctly	Pass	None needed
2	Registration password match	Password: "pass123", Confirm: "pass456"	Error saying passwords don't match	Error appeared	Pass	None needed
3	Registration email format	Email: " testemail.com " (missing @)	Error saying email is invalid	Error appeared	Pass	None needed
4	Registration short password	Password: "123" (less than 8 chars)	Error saying password too short	Error appeared	Pass	None needed
5	Registration duplicate email	Email already in database	Error saying email already registered	Error appeared	Pass	None needed
6	Registration producer role	Select "Producer", fill business name	Account created with producer role	Account created correctly	Pass	None needed
7	Login with wrong password	Existing email, wrong password	Error message, no login	Stayed on login page	Pass	None needed
8	Login with wrong email	Email not in database, any password	Error message, no login	Error appeared	Pass	None needed

9	Login with empty fields	Empty email and password	Error message	Error appeared	Pass	None needed
10	Login with correct details	Valid email and password	Redirect to home page	Logged in successfully	Pass	None needed
11	Products page loading	Visit /products page	Display products from database	Showed "0 of 0 products"	Fail	Fixed API endpoint and port number
12	Products page after fix	Visit /products page	Display 12 products	Products showed correctly	Pass	None needed
13	Category filter - Vegetables	Click "Vegetables" button	Show only vegetable products	Only vegetables showed	Pass	None needed
14	Category filter - Bakery	Click "Bakery" button	Show only bakery products	Only bakery showed	Pass	None needed
15	Category filter - All	Click "All" button	Show all products	All products showed	Pass	None needed
16	Search function	Type "sourdough" in search bar	Show sourdough products	Sourdough products showed	Pass	None needed
17	Search function no results	Type "pizza" in search bar	Show "No products found"	No products message appeared	Pass	None needed
18	Sort by price low to high	Select "Price: Low to High"	Products sorted by cheapest first	Sorted correctly	Pass	None needed
19	Sort by price high to low	Select "Price: High to Low"	Products sorted by most	Sorted correctly	Pass	None needed

			expensive first			
20	Add to basket	Click "Add to Cart" on product	Item appears in basket	Basket showed item	Pass	None needed
21	Update basket quantity	Click + button on basket item	Quantity increases by 1	Quantity increased	Pass	None needed
22	Remove from basket	Click "Remove" button	Item removed from basket	Item disappeared	Pass	None needed
23	Basket total calculation	Add 2 items at different prices	Total adds up correctly	Total was correct	Pass	None needed
24	Checkout stock deduction	Order 5 items where stock was 8	Stock becomes 3	Stock decreased to 3	Pass	None needed
25	Checkout insufficient stock	Order 10 items where stock was 3	Error, order not placed	Insufficient stock error	Pass	None needed
26	Checkout delivery option	Select "Home Delivery"	Delivery fee £3.50 added	Fee added to total	Pass	None needed
27	Checkout collection option	Select "Click & Collect"	No delivery fee	No fee added	Pass	None needed
28	Loyalty points after order	Place £12.80 order	Earn 128 points	Showed 0 points	Fail	Added points calculation to order endpoint
29	Loyalty points after fix	Place £12.80 order	Earn 128 points	Points showed correctly	Pass	None needed

30	Loyalty points redemption	Apply 100 points at checkout	£5 discount applied	Discount worked	Pass	None needed
31	Loyalty points insufficient	Try to redeem 250 points with 100	Button disabled, shows points needed	Button disabled correctly	Pass	None needed
32	Producer login	Producer credentials	Redirect to Dashboard	Dashboard loaded	Pass	None needed
33	Producer sees own products	New producer logs in	Empty product list	Saw other producers' products	Fail	Added producer_id filter to product query
34	Producer sees own products after fix	New producer logs in	Empty product list	Showed "No products yet"	Pass	None needed
35	Producer add new product	Fill form, click submit	Product saved to database	Product appeared in dashboard	Pass	None needed
36	Producer update stock	Change stock from 20 to 15	Stock updates in database	Dashboard showed new stock	Pass	None needed
37	Producer update price	Change price from £3.20 to £3.50	Price updates in database	Price changed correctly	Pass	None needed
38	Producer view orders	Click "View Orders" tab	Show orders for producer's products	Orders displayed	Pass	None needed
39	Producer update order status	Change status to "Ready"	Status updates	Status changed	Pass	None needed

40	Navigation bar after login	Customer logs in	Shows "Logout"	Still showed "Login/Register"	Fail	Fixed AuthContext to re-render navbar
41	Navigation bar after fix	Customer logs in	Shows "Logout"	Logout showed correctly	Pass	None needed
42	Account page order history	Customer with orders logs in	Display past orders	Order history showed	Pass	None needed
43	Account page no orders	New customer logs in	Shows "No orders yet"	Message appeared	Pass	None needed
44	Map API loading	Visit About page	Map shows with 4 markers	Map loaded correctly	Pass	None needed
45	Map marker click	Click on farm marker	Popup shows farm details	Popup appeared	Pass	None needed
46	Dark mode toggle	Click dark mode button	Theme changes across site	Dark mode applied	Pass	None needed
47	Dark mode persistence	Refresh page after toggle	Dark mode stays on	Setting saved	Pass	None needed
48	Footer links - Products	Click "Browse Products"	Navigate to /products	Products page loaded	Pass	None needed
49	Footer links - About	Click "About Us"	Navigate to /about	About page loaded	Pass	None needed
50	Logout button	Click "Logout"	Clear session, redirect to home	Logged out successfully	Pass	None needed

